

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
# Lab 03 Setup
library(tidyverse)
library(lubridate)
library(here)
library(dplyr)
library(ggplot2)

getwd()
```

```
## [1] "/home/guest/R/RemoteRepositoryClone"
```

```
ecotox_data <- read.csv(file = here("./Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv"))
NEON_data <- read.csv(file = here("./Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv"),
                      stringsAsFactors = TRUE)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Neonicotinoids are the most popular insecticide class in the United States. While it is effective in killing pests to agriculture, it also kills important insects such as butterflies and bees that are crucial to pollination and other vital agricultural processes. Looking closer at how this insecticide affects insects would allow for a greater understanding of how we can protect pollinators and other important insects while effectively killing insects that harm crop production.

Source: <https://www.nrdc.org/stories/neonicotinoids-101-effects-humans-and-bees>

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Studying litter and woody debris is important because they are both essential components in forest ecosystems as they contribute to nutrient cycles, provide habitats for many forest animals, provide food, and support new tree growth. Currently human clearance of litter and woody debris compromises many of these benefits which has elevated forest degradation and species endangerment.

Source: <https://www.nrdc.org/bio/julee-boan/removing-forest-waste-degrades-ecosystem-structure-and-function>

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Litter is dropped from the forest canopy and is sampled from PVC basket traps elevated around 80 cm off the ground. 2. Fine wood debris dropped from the forest canopy is sampled from rectangular ground traps. 3. Litter and debris sampling occurs at sites where woody vegetation exceeds 2m in height.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
# Dataset Dimensions
dim(ecotox_data)
```

```
## [1] 4623 30
```

```
# The first element of dim() is the number of rows, the second element is the number of columns  
# The ecotox_data set is 4623 rows and 30 columns
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
# Summary of Neonicotinoids Effects
```

```
summary(ecotox_data$Effect)
```

```
##      Length      Class      Mode  
##      4623 character character
```

```
#sort(ecotox_data$Effect, decreasing=FALSE)
```

```
names(sort(table(ecotox_data$Effect), decreasing=T))
```

```
## [1] "Population"      "Mortality"        "Behavior"         "Feeding behavior"  
## [5] "Reproduction"    "Development"      "Avoidance"        "Genetics"  
## [9] "Enzyme(s)"       "Growth"           "Morphology"       "Immunological"  
## [13] "Accumulation"    "Intoxication"     "Biochemistry"     "Cell(s)"  
## [17] "Physiology"      "Histology"        "Hormone(s)"
```

```
names(which.max(table(ecotox_data$Effect)))
```

```
## [1] "Population"
```

Answer: The dimensions of the neonicotinoids ecotoxicology data set is 4623 rows by 30 columns. The most common effect studied is Population, followed by Mortality and Behavior. These would be the greatest interest because Population would show how the use of neonicotinoids affect insect populations (i.e. growth), mortality would return the number of population deaths that occur as a result of the insecticide, and behavior would return data about eating habits, communication, and reproduction changes that arise after insecticide use.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
# Summary of Studied Species
```

```
summary(as.factor(ecotox_data$Species.Common.Name), maxsum=7)
```

```
##      Honey Bee      Parasitic Wasp Buff Tailed Bumblebee  
##      667          285          183  
## Carniolan Honey Bee      Bumble Bee      Italian Honeybee  
##      152          140          113  
##      (Other)  
##      3083
```

Answer: The six most commonly studied species in the dataset are Honey Bees, Parasitic Wasps, Buff Tailed Bumblebees, Carniolan Honey Bees, Bumble Bees, and Italian Honeybees. The honeybees and bumblebees in the top six studied species are important pollinators for crop production, and Parasitic Wasps, while reliant on deaths of other insects to repopulate, are also important pollinators. These species in particular are at risk from neonicotinoids because pollen becomes toxic after the insecticide is absorbed by the plant, meaning these species die or experience long-lasting damage after ingesting nectar or pollen.

Source: https://xerces.org/sites/default/files/2018-05/16-023_01_XercesSoc_ExecSummary_How-Neonicotinoids-Can-Kill-Bees_web.pdf

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
# Col.1..Author Data
```

```
class(ecotox_data$Conc.1..Author.)
```

```
## [1] "character"
```

```
head(ecotox_data$Conc.1..Author., n=25)
```

```
## [1] "27.2" "19.7" "47" "25" "13" "268" "170" "28"
## [9] "48" "40" "83" "900" "15.3" "20.4" "5" "5"
## [17] "NR" "~10" "65.56" "635.4" "239.1" "NR/" "80.66" "1075.7"
## [25] "75.27"
```

```
tail(ecotox_data$Conc.1..Author., n=25)
```

```
## [1] "0.40/" "0.40/" "0.40/" "0.40/" "0.40/" "0.40/" "2.4"
## [8] "2.4" "3.0/" "6" "224.05/" "224.05/" "224.05/" "0.15/"
## [15] "0.15/" "75/" "0.048/" "0.45/" "0.45/" "0.28/" "0.28/"
## [22] "0.0113" "3.4859" "6/" "6"
```

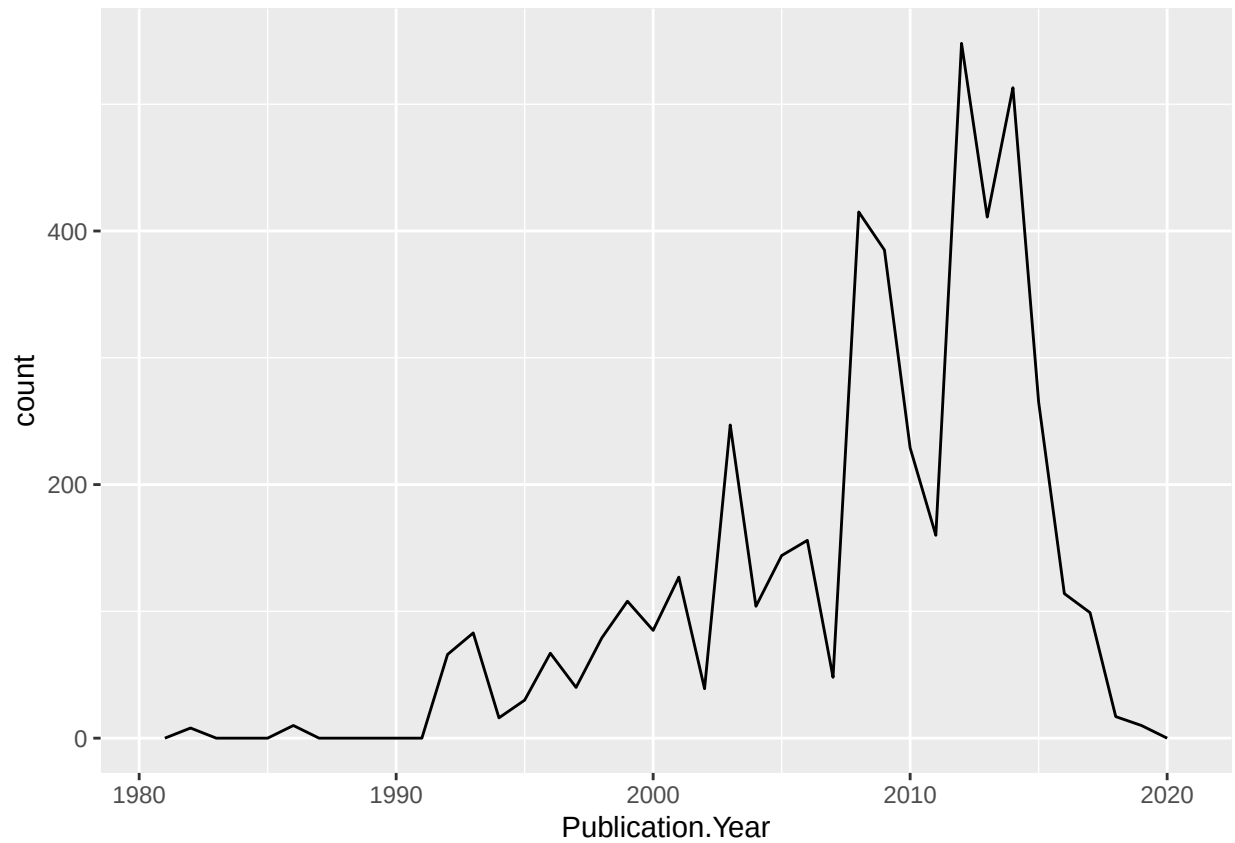
Answer: The elements of the `Conc.1..Author` column are characters, not numbers. Looking at the data in the column, concentrations are listed in different formats from numeric, to estimations, or having no concentration listed. Given the different results, the best format to present the data is in character class.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
# Publication Year Plot
```

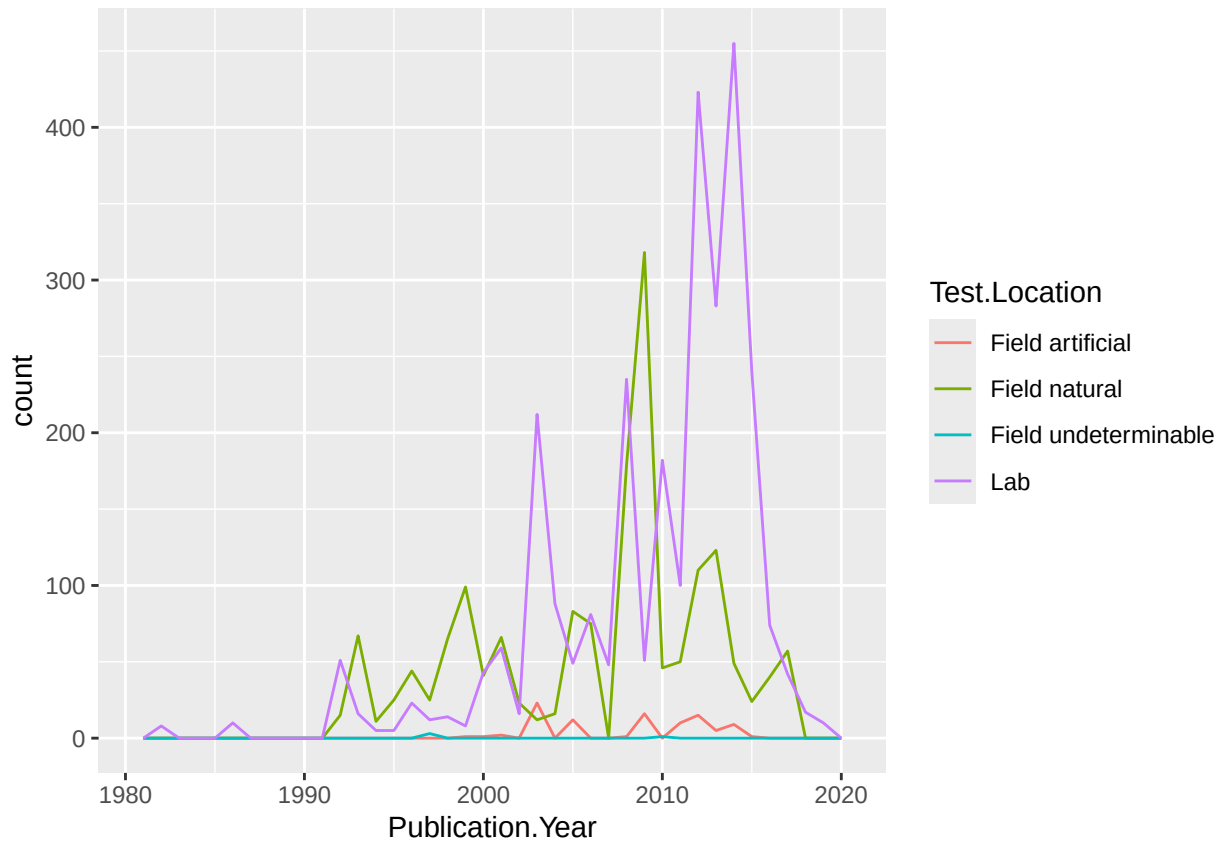
```
ggplot(data = ecotox_data, aes(x = Publication.Year)) +
  geom_freqpoly(binwidth = 1)
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
# Plot of Test Locations in Publication Years
```

```
ggplot(data = ecotox_data, aes(x=Publication.Year, color = Test.Location)) +  
  geom_freqpoly(binwidth = 1)
```



Interpret this graph. What are the most common test locations, and do they differ over time?

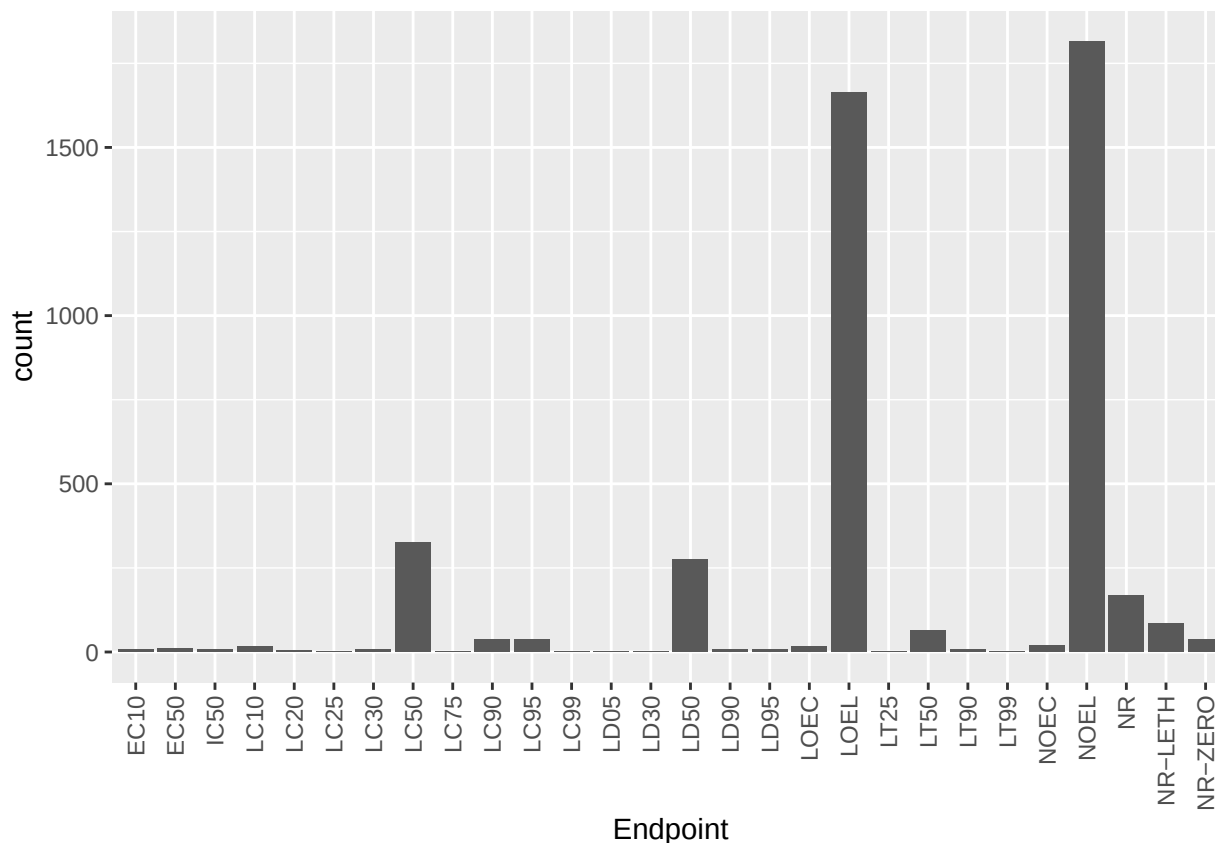
Answer: Looking at the graph, the most common test locations are the lab and in natural field locations. Over time, testing in the lab remained higher than the rest, but peaked largely between the 2010-2015 publication years. Just before 2010, testing in the natural field rose to higher rates than testing in the lab.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[TIP: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
# Plot of Endpoint counts

ggplot(data = ecotox_data, aes(x=Endpoint)) +
  geom_bar(stat = "count") + theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



Answer: The two most common end points are NOEL and LOEL. NOEL is defined as no-observable-effect-level in a terrestrial database while LOEL is defined as the lowest-observable-effect-level in a terrestrial database.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
#collectDate Data and Reformatting
```

```
#class(NEON_data$collectDate)
```

```
NEON_data$collectDate <- as.Date(NEON_data$collectDate, format = "%Y-%m-%d")
class(NEON_data$collectDate)
```

```
## [1] "Date"
```

```
aug_2018_dates <- unique(
  NEON_data$collectDate[
    format(NEON_data$collectDate, "%Y-%m") == "2018-08"
  ]
)
aug_2018_dates
```

```
## [1] "2018-08-02" "2018-08-30"
```

```
# Litter was sampled on the 2nd and 30th of August, 2018.
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
#plotID Data
```

```
unique(NEON_data$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

```
summary(NEON_data$plotID)
```

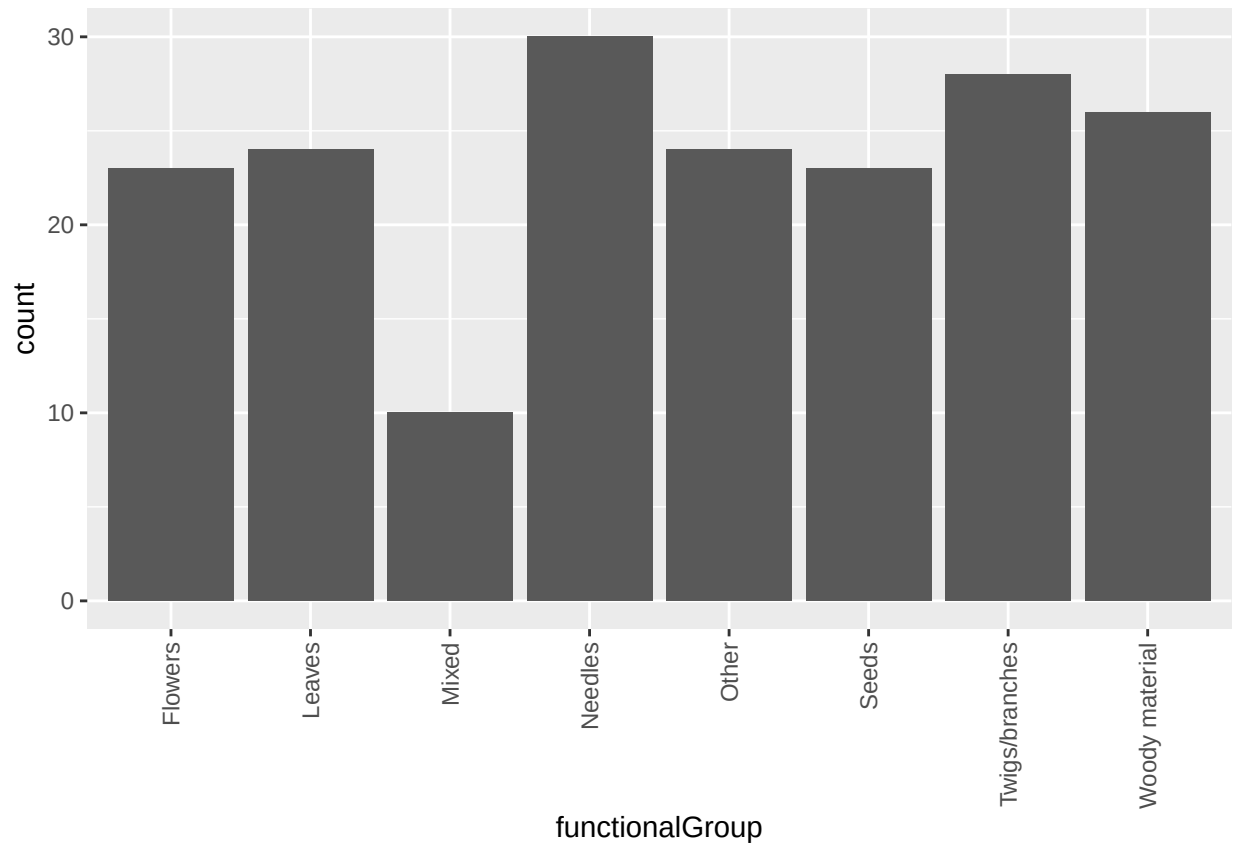
```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061  
##      20      19      18      15      14      8      16      17  
## NIWO_062 NIWO_063 NIWO_064 NIWO_067  
##      14      14      16      17
```

Answer: `unique()` returns a list of each unique plotID found in the dataset, while `summary()` returns a list of the different plotIDs in the data set and how many times the plotID is listed (frequency).

14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
# Plot of functional groups by count
```

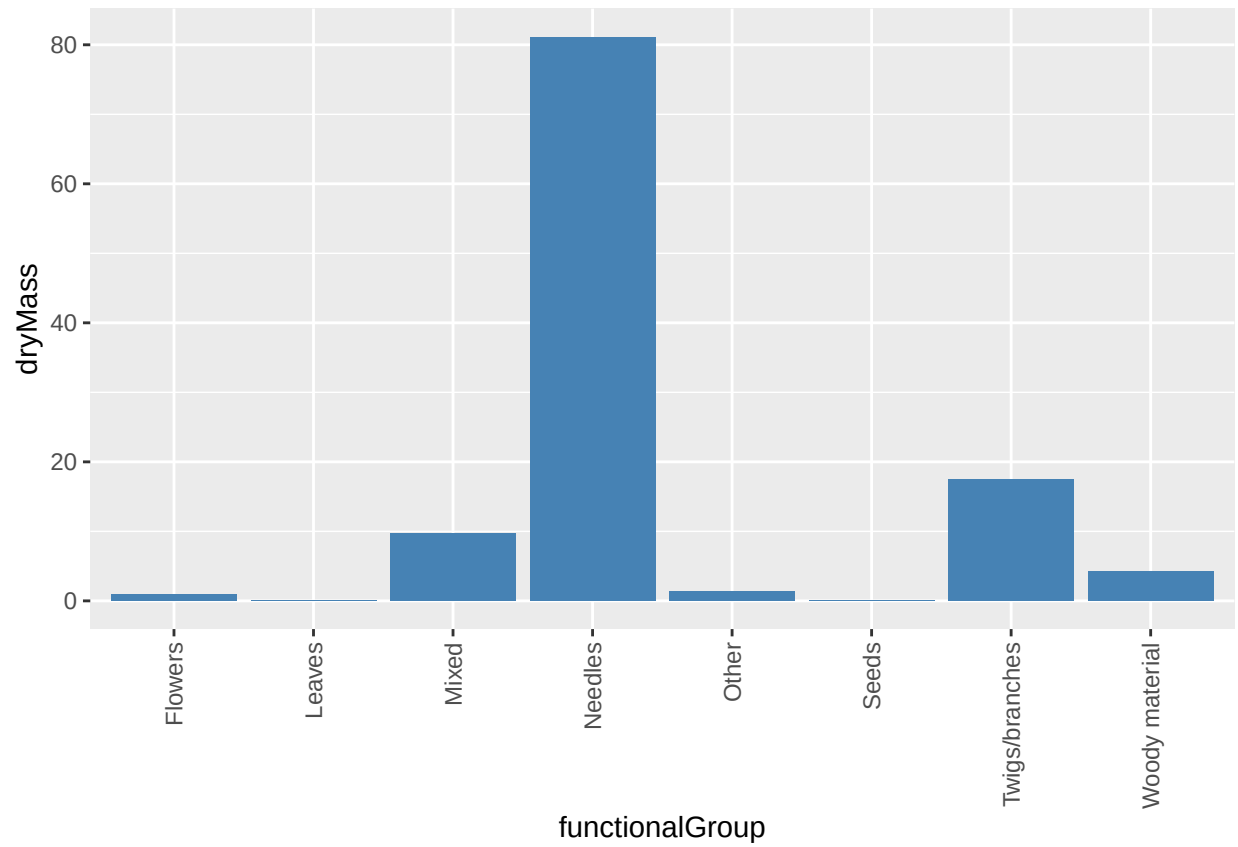
```
ggplot(data = NEON_data, aes(x=functionalGroup)) + geom_bar(stat = "count") +  
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



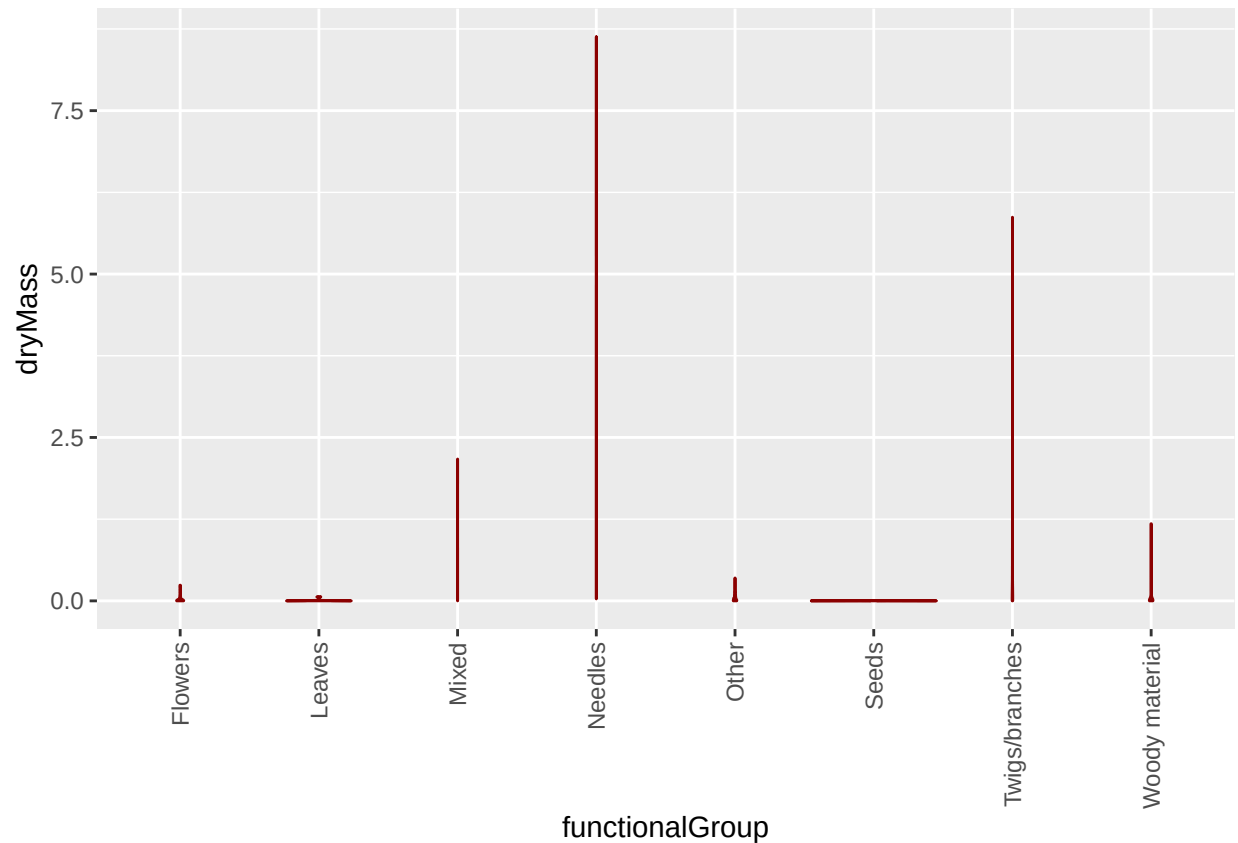
15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
# Box plot and violin plot of dry mass per functional group

ggplot(data = NEON_data, aes(x=functionalGroup, y=dryMass)) + geom_bar(stat="identity",
  fill = "steelblue") + theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



```
ggplot(data = NEON_data, aes(x=functionalGroup, y=dryMass)) + geom_violin(color = "darkred") +  
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: In this case, the boxplot provides a more effective visualization because it shows the dry mass per functional group which uses categorical separation to easily display comparable dry mass values between each functional group. Violin plots would be used to show distribution of data within a functional group, but since the data being graphed is already split into separate groups and is only plotting dry mass, there is no horizontal distribution on the graph resulting in the less effective thin-line visualization.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: The types of litter with the highest biomass are needles, twigs/branches, and mixed litter.